

Project Synopsis
On
StockTracker - (Inventory Management System)

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1. Title

StockTracker - (Inventory Management System)

2. Introduction

Inventory management is a critical function for businesses of all sizes. Keeping track of stock levels, monitoring sales, managing suppliers, and generating reports are essential activities that directly impact profitability. Traditional methods—such as paper-based registers or basic spreadsheets—are prone to human error, inefficiency, and a lack of real-time visibility.

StockTracker is a comprehensive, web-based inventory management system designed to automate the stock management workflow. Built using modern web technologies including HTML5, CSS3, PHP 8.x, and PostgreSQL, the system provides a feature-rich dashboard, product management, and automated invoicing. The significance of this project lies in its practical approach to solving real-world challenges faced by small and medium enterprises (SMEs) by providing a centralized digital platform.

3. Objectives

- To design and develop a functional web-based inventory system with real-time tracking.
- To implement complete CRUD (Create, Read, Update, Delete) operations for products and suppliers.
- To develop a sales and invoicing module that supports automated invoice generation.
- To build an interactive dashboard with charts and visual analytics for quick decision-making.
- To create a Point of Sale (POS) interface for fast, counter-level billing operations.
- To support CSV import/export for bulk data handling.
- To generate automated low-stock alerts and ensure system security using prepared statements.

4. Scope

4.1 Included in Scope

- **Product Inventory:** Categorization, pricing, and stock level indicators (low, medium, high).
- **Sales Management:** Multi-item invoicing, discount application, and printable receipts.
- **Supplier Database:** Full CRUD operations and purchase history tracking.
- **Dashboard:** KPI metrics and sales charts using Chart.js.
- **Quick POS Module:** Rapid counter sales interface.
- **Reporting:** Profit and loss analysis and stock movement tracking.

4.2 Excluded from Scope

- Multi-branch or warehouse management across different geographical locations.
- Integration with third-party payment gateways (e.g., Stripe, PayPal).
- Mobile native application development (the system is responsive web-based only).

5. Methodology

The project follows a modular development methodology combining Agile principles with a classic three-tier web architecture.

5.1 Architecture

- **Frontend (Presentation Tier):** HTML5, CSS3 with Bootstrap 5 for responsive styling.
- **Backend (Logic Tier):** PHP 8.x utilizing PDO for secure database interaction.
- **Database (Data Tier):** PostgreSQL for persistent data storage.

5.2 Tools and Technologies

Component	Technology
Frontend Structure	HTML5, CSS3, Bootstrap 5

Client-Side Logic	JavaScript (ES6+)
Backend Language	PHP 8.x
Database	PostgreSQL
Version Control	Git

6. Literature Review

Inventory systems have evolved from manual record-keeping to sophisticated cloud-based solutions. Early computerized systems were limited by platform dependency. Today, web technologies allow for real-time tracking and remote access. Research in web-based applications highlights the effectiveness of the MVC (Model-View-Controller) architecture for improving software maintainability (Leff & Rayfield, 2001). This project builds upon these established frameworks to deliver a cost-effective, open-source solution tailored for small business use and academic demonstration.

7. Implementation Plan

Phase	Tasks	Duration
Phase 1: Planning	Requirement analysis and database schema design	Week 1
Phase 2: Core Dev	Product CRUD and dashboard KPI setup	Week 2–3
Phase 3: Business	Sales, invoicing, and supplier management	Week 4–5
Phase 4: Advanced	POS module, reports, and CSV tools	Week 6–7
Phase 5: Testing	Bug fixing, documentation, and final deployment	Week 8

8. Expected Outcomes

- A fully functional web-based system accessible via any modern browser.
- A real-time dashboard displaying key performance indicators and interactive charts.
- An automated sales system with printable invoices and transaction history.
- Automated low-stock alerts to prevent inventory shortages.
- Secure data handling to prevent SQL injection and unauthorized access.

9. Conclusion

StockTracker addresses the need for an affordable, easy-to-use inventory management system. By leveraging modern technologies like PHP and PostgreSQL, the project demonstrates the practical application of full-stack development in solving business problems. The system covers the entire lifecycle from stock monitoring to financial reporting, making it a valuable tool for small business owners.

10. References

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